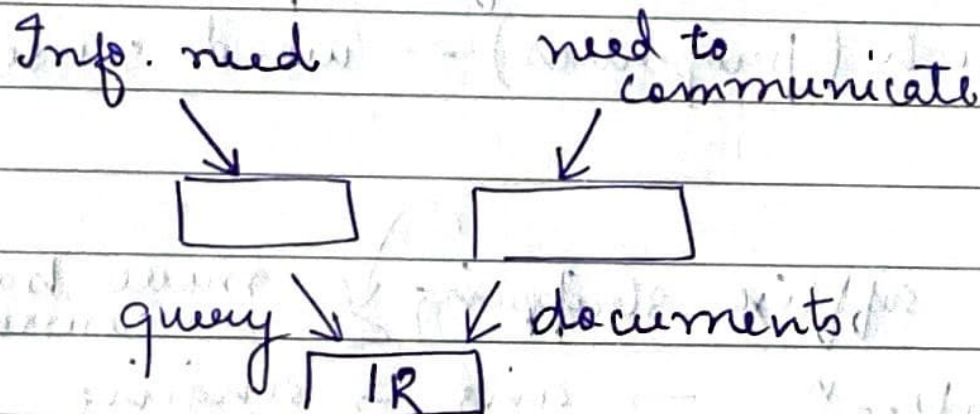


23rd Jan 26 / CS6370

⇒ IR → Applicaⁿ



- Structural & functional correspondence is impot.
- HCI problem have to be solved for IR. It is currently assumed, we know what info. we need.
- Search engine is an IR sys. Used to info from text or document.

Doc 1 Doc 5 {1 sentence per doc}

analogy

{ letters $\leftrightarrow$ electrons
words $\leftrightarrow$ atoms
sentence $\leftrightarrow$ molecules

why?

atomic
units of
meaning

Tokenize your text

there are
some
structural
constraints

this will
lead to

formaⁿ of
BOW (Bag of
words)

like that
few sentences are
syntactically correct; some are
incorrect.

few more pre-processing steps

(spell check, suffix stripping, stopword
removal, etc)

here we
are looking
at the structure
of words

no. of letters matches
the diff. of
morphology of Sanskrit
vs English.

- morphology (structure) → fixes spell checker, suffix clipping
- parts of speech (properties) → word sense
- phonetics
- semantics

suffix stripping → move back to more
canonical form

in English { inflecⁿ → sing → singing (verb → verb)
derivaⁿ → educate → education (verb → noun)

- IR → Knowledge light task. We req. very
less knowledge to solve the problem.
- Stopword Removal - remove the
stopwords (most common
words used in the language) it is not
universal

- funcⁿ words → they glue the words together, but themselves have not much value.
//_

Russell & Norwing Ch-1. Mandatory Reading
(AI Text book)

✓ Stemming vs Lemmatization → tries to reach the correct root
(drawing → draw) (eg. dancing → dance)
direct removal

✓ Porter's Stemmer

- Stemming → chopping off word endings using simple rules. It does not care anything about grammar.

→ eg. removes → ing, id, s

studied → studi } these two are
caring → car } not even
↓
actual meaning got distorted.

useful → search engines, large-scale text mining
speed is preferred over linguistic accuracy.

- Lemmatization → reduces a word to its dictionary base form (lemma) using vocab, grammar, parts-of-speech info.

studied → study

useful → NLP pipelines, chatbots, QA, sentiment analysis

- Porter's Stemmer → earliest & most widely used stemming algo. It is a rule-based algo. that reduces words to their stem by iteratively removing 'common eng' suffixes.

Each step →

1. looks for a specific suffix pattern
2. checks for a linguistic condition
3. removes or replaces the suffix

Pros → fast, lightweight, no dictionary, works for IR engine, lang.-agnostic.

Cons → can prod. non-words, not context-aware, sometimes cause over- or under-stem, has only english specific rules.